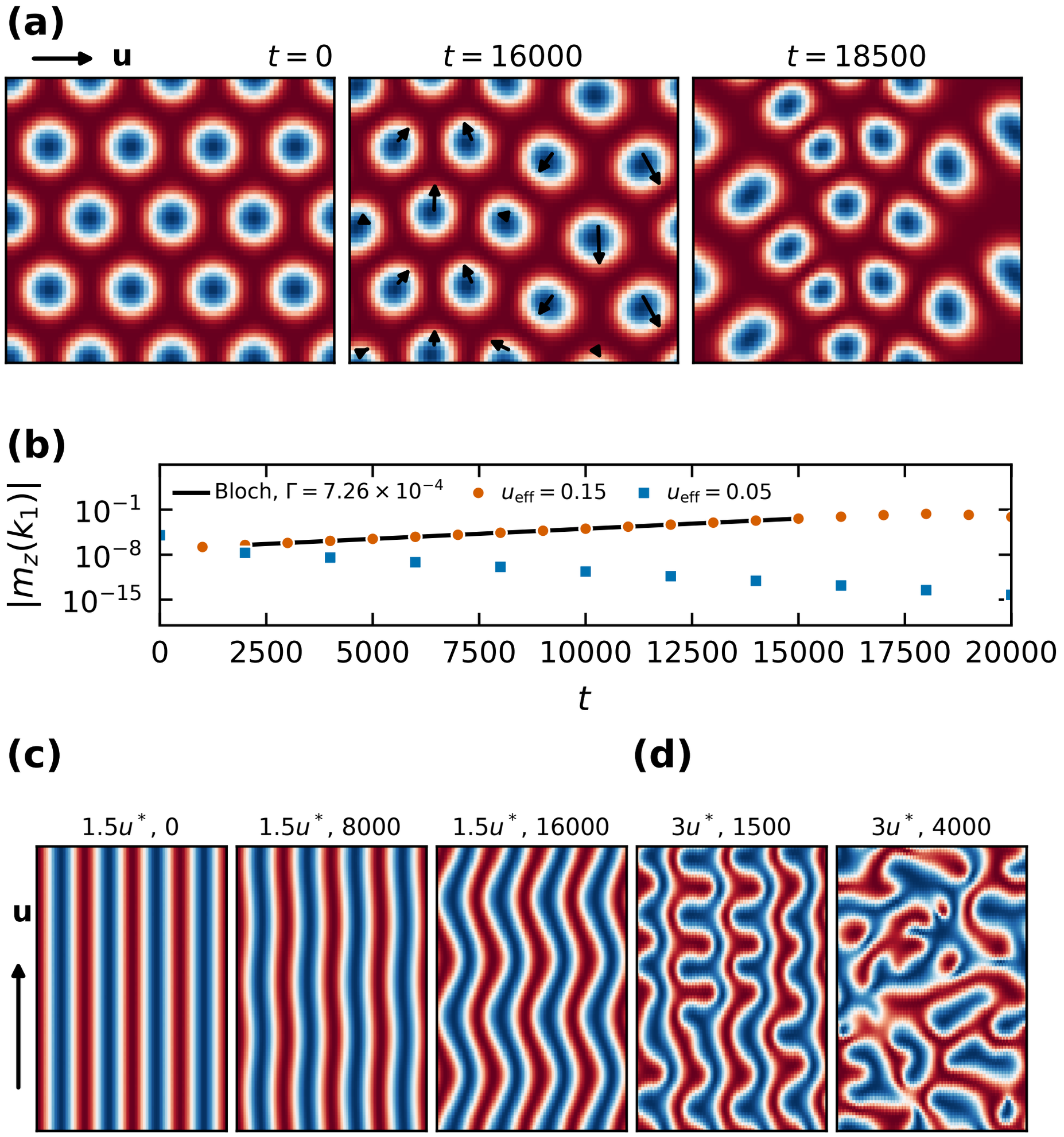

(a)
u
$t = 0$
$t = 16000$
$t = 18500$
(b)
$|m_z(k_1)|$
Bloch, $\Gamma = 7.26 \times 10^{-4}$
$u_{\mathrm{eff}} = 0.15$
$u_{\mathrm{eff}} = 0.05$
10^{-1}
10^{-8}
10^{-15}
0
2500
5000
7500
10000
12500
15000
17500
20000
t
(c)
(d)
$1.5u^*$, 0
$1.5u^*$, 8000
$1.5u^*$, 16000
$3u^*$, 1500
$3u^*$, 4000
u